

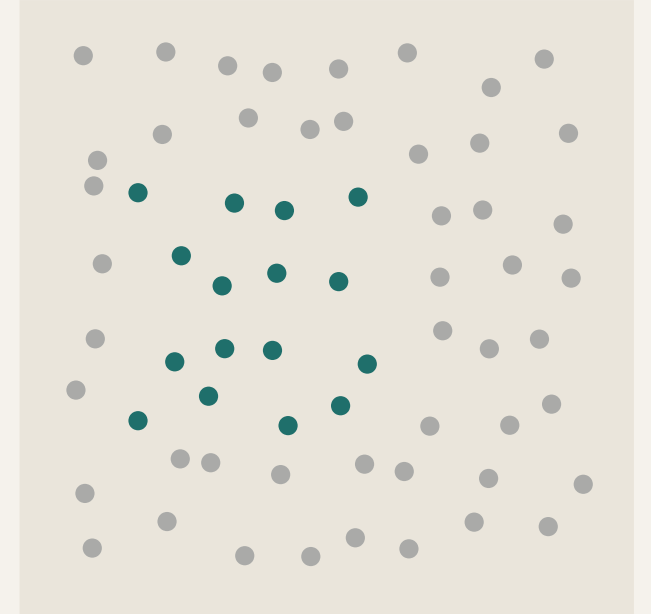
Fashion search: what the literature says

Recent papers mapped to our agenda

Compiled semantic predicates
RSA

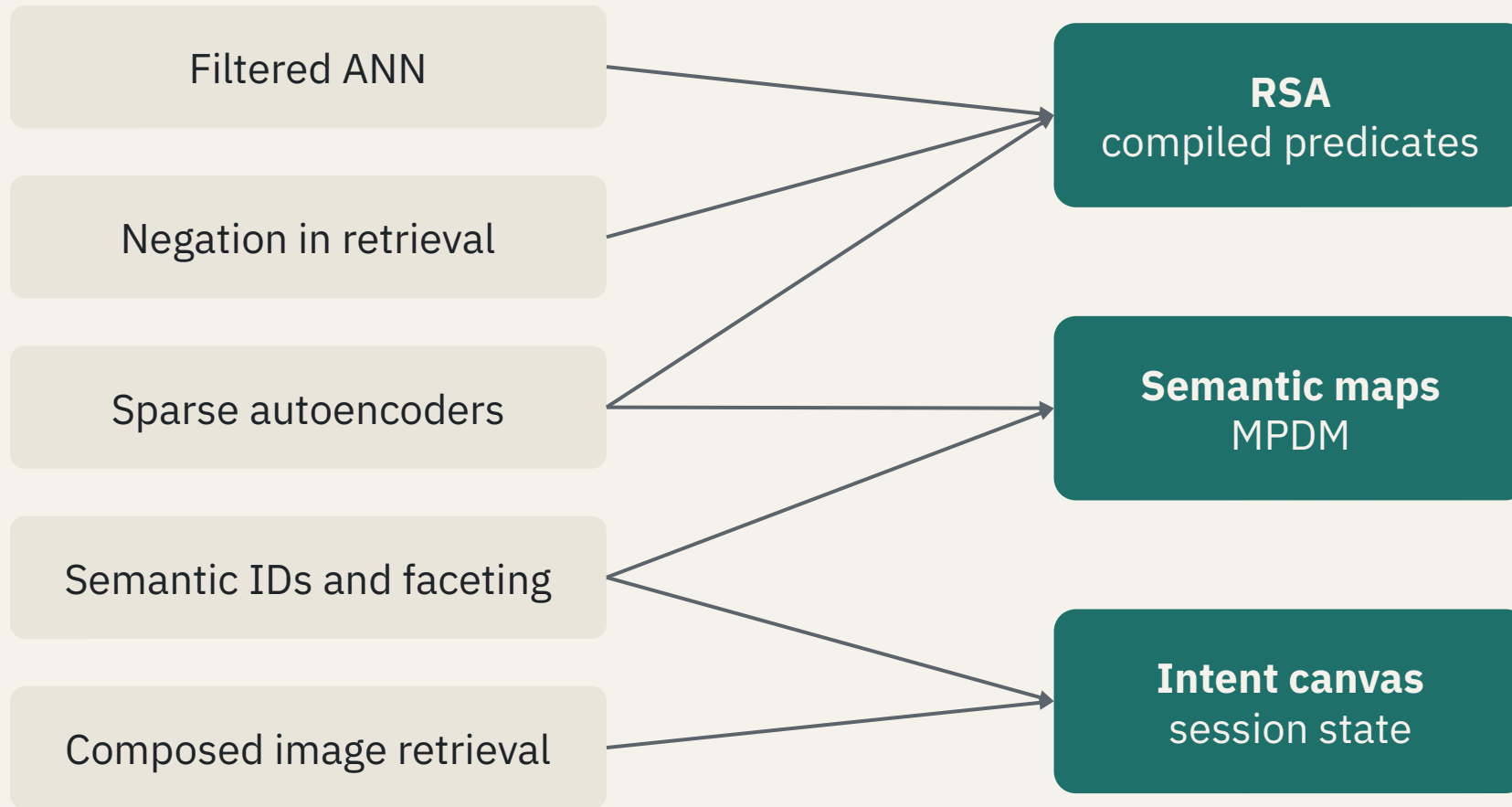
Semantic maps
MPDM

Intent canvas
Session state



formal – casual × minimal – ornate

Five research fronts feed our three ideas

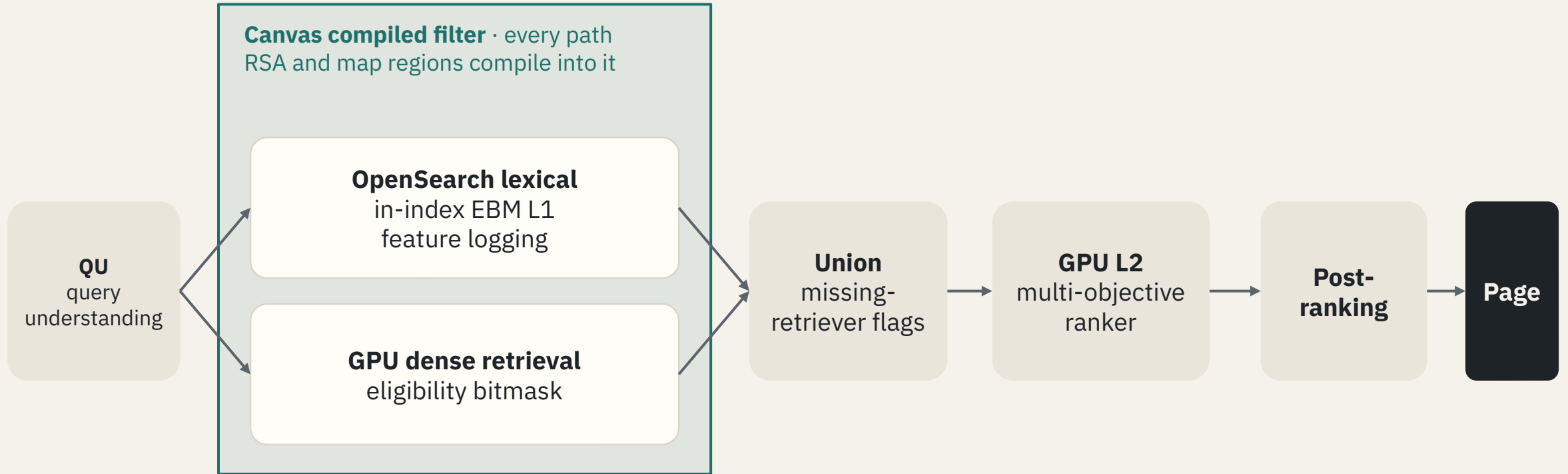


RSA draws on three fronts: filtered ANN, negation and SAEs.

Maps borrow from SAEs and semantic IDs.

Canvas borrows from semantic IDs and composed retrieval.

Where each idea plugs into our search stack



The canvas compiles once per turn and constrains **every** retriever, lexical and dense alike. Missing-retriever flags let L2 score items that only one path found.

Filtered search is fast now; the predicate is still assumed

[Figure: ACORN, predicate-subgraph traversal / neighbor expansion, or recall vs. selectivity]

Placeholder: PDF not reachable at build time. Patel et al. (2024), arXiv:2403.04871.

OUR LIVE TRAVERSAL RULE (RSA INSIDE HNSW)

navigate by $s_{\text{dense}}(q, x)$, admit x if $S_{\text{semantic}}(x) \geq \tau$

ACORN makes HNSW predicate-agnostic by traversing the predicate subgraph.

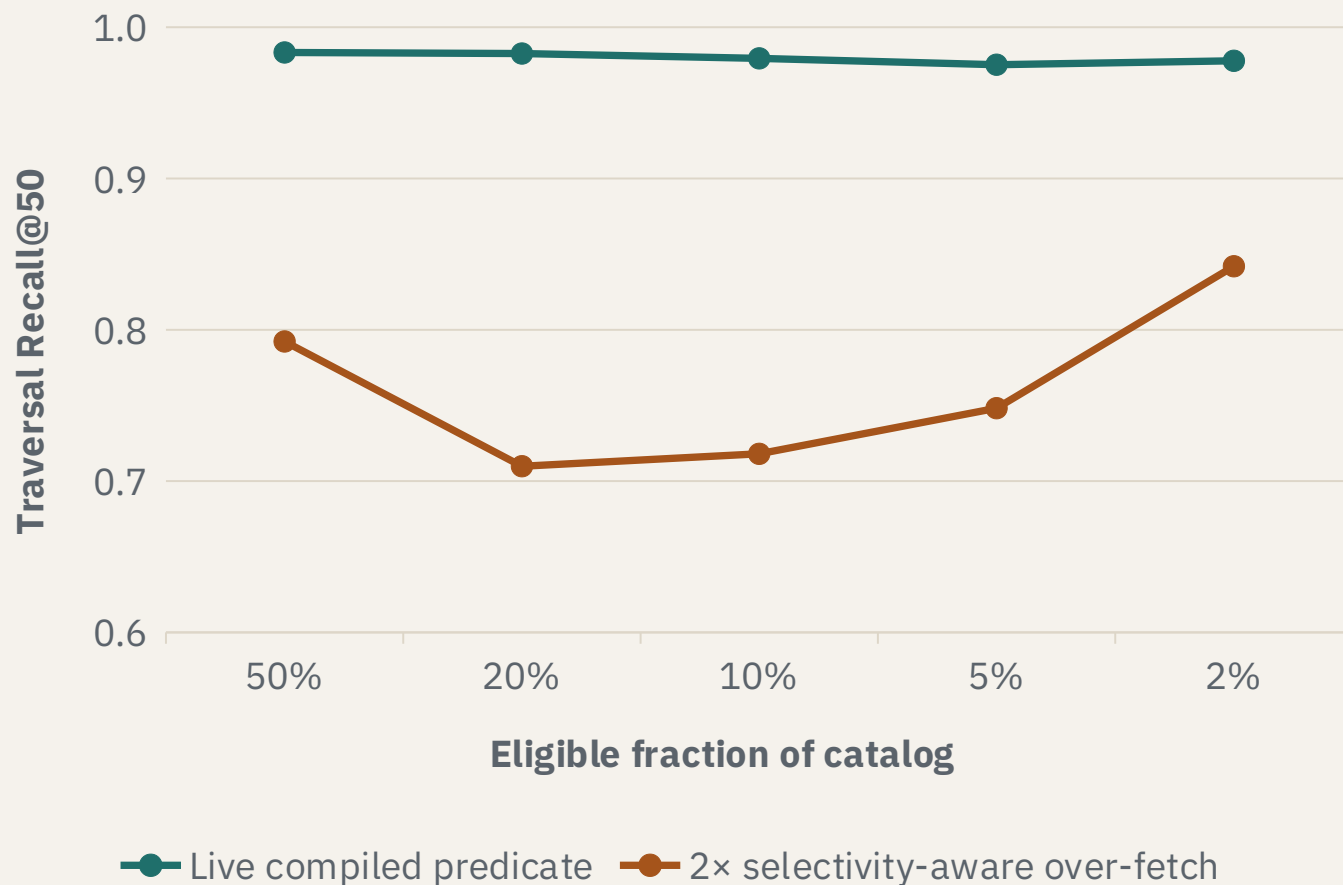
FAVOR: 1.3–5× higher QPS at Recall@10 = 95%.

EMA: 1.68–12.25× speedup over general-filtering ANN methods.

Our read: all three take the predicate as given. A compiled RSA program can be that predicate.

OUR RESULT IN CONTEXT

Live predicates hold recall as eligibility shrinks



AT 2% ELIGIBILITY

0.978 vs **0.842**

Recall@50, live vs 2x over-fetch

14.57 ms vs **20.14 ms**

No tested over-fetch budget got within 0.005 of live recall, in 15 of 15 conditions.

Next: measure local, not global, selectivity (anticorrelated queries).

RSA compiles a concept into a ~216-byte program

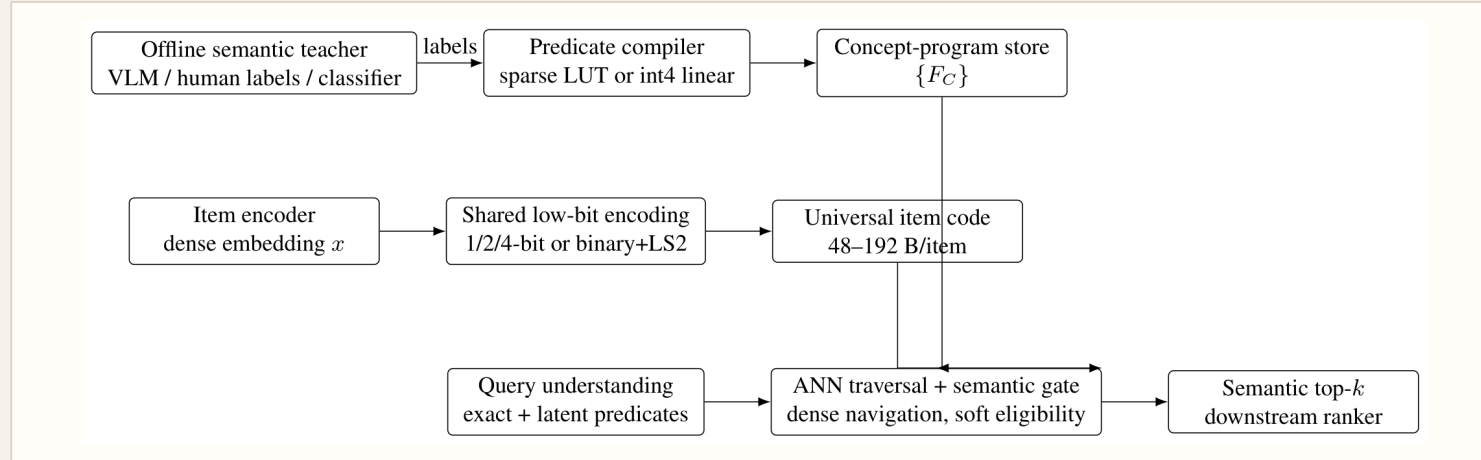


Figure 1 from Al-Shater (2026), local PDF, p. 2. License: author's own.

$$p_C(x) = \sigma(a'_C F_C(x) + c_C) \quad M(N, K) = N B_{\text{item}} + K B_{\text{program}}$$

$$S_Q(x) = \sum_{C \in Q^+} \log p_C(x) + \sum_{C \in Q^-} \log(1 - p_C(x))$$

$$P_C(x) = a_C n_+(x) + \Delta_C \sum_{t=0}^3 2^t \text{popcnt}(B_x \wedge M_{Ct})$$

56 B

per item: shared binary code

~216 B

per predicate program

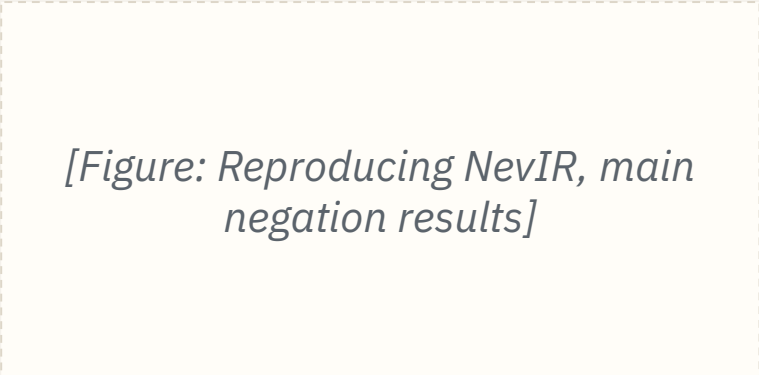
0.337

recall at 20% retention · dense 0.207
· PQ64 0.352 · FP32 0.363

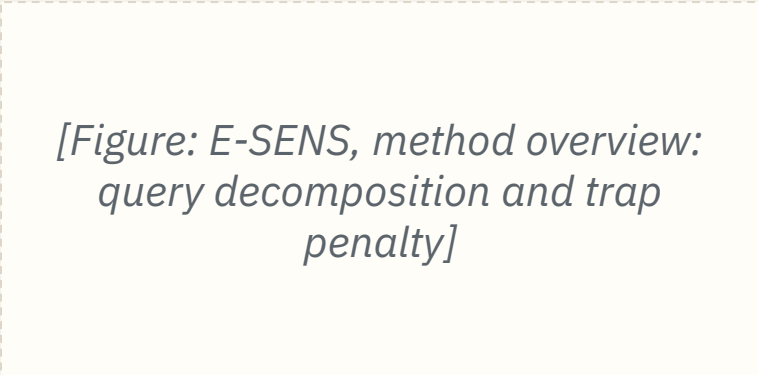
96–134 ns

per predicate invocation inside HNSW

Negation is still broken; E-SENS is the bar to beat



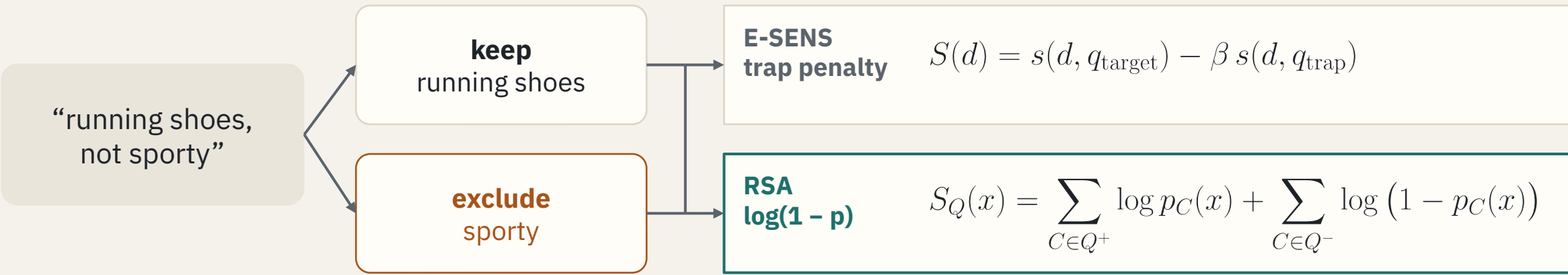
Placeholder: PDF not reachable at build time. van den Elsen et al. (2025), arXiv:2502.13506.



Placeholder: PDF not reachable at build time. Kim et al. (2026), arXiv:2608.30130.

Most retrievers still rank negated pairs at or below random.

Our bar is now **beating E-SENS**, not zero-shot directions.



SAEs could hand RSA a concept vocabulary for free

[Figure: SAE embeddings toolkit, Figure 1: converting documents into interpretable SAE embeddings]

Placeholder: PDF not reachable at build time. Jiang et al. (2025), arXiv:2512.10092.

SPARSE AUTOENCODER

$$z = \text{ReLU}(W_e x + b_e), \quad \mathcal{L} = \|x - W_d z\|_2^2 + \lambda \|z\|_1$$

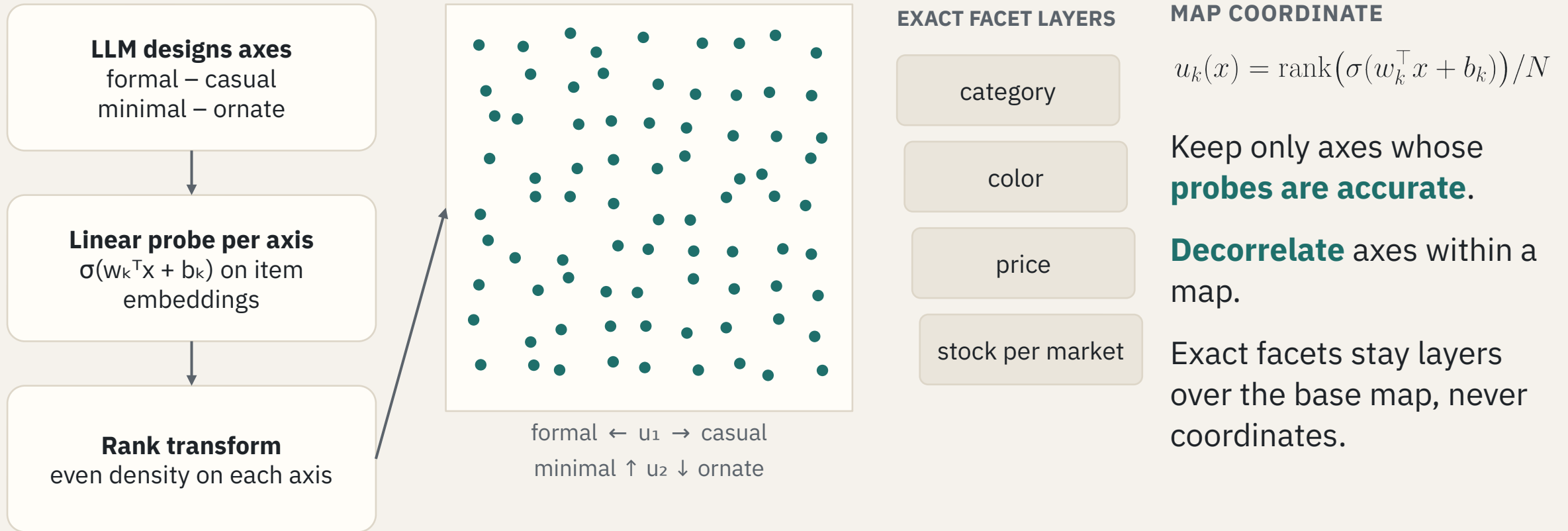
SAE features give RSA a **teacher-free** concept vocabulary.

Jiang et al. use SAE embeddings for data diffing, correlations, clustering and retrieval.

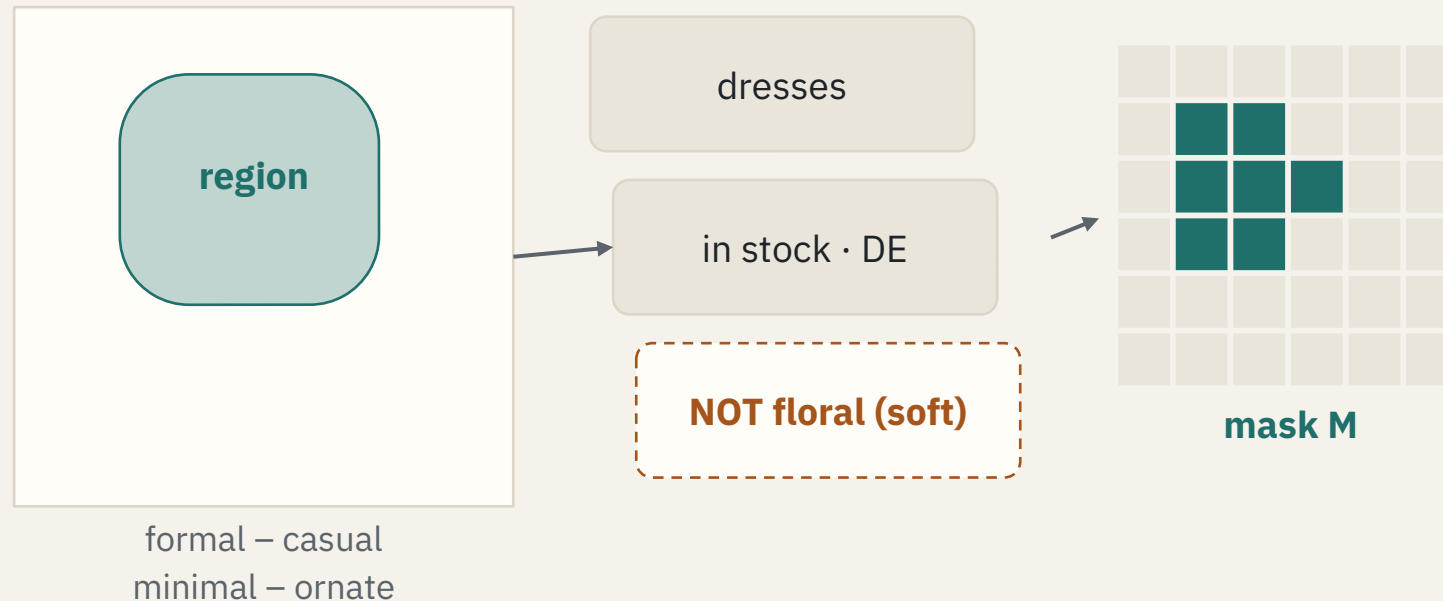
Park et al. (CL-SR) turn SAE latents into concept-level sparse retrieval.

Risks we must test: feature absorption, domain shift, extra compute.

An LLM designs the axes; probes place the items



Each clarification becomes layer algebra



CANVAS MASK AS LAYER ALGEBRA

$$M = R_{\text{base}} \wedge L_{\text{dress}} \wedge L_{\text{stock,DE}} \wedge \neg L_{\text{floral}}$$

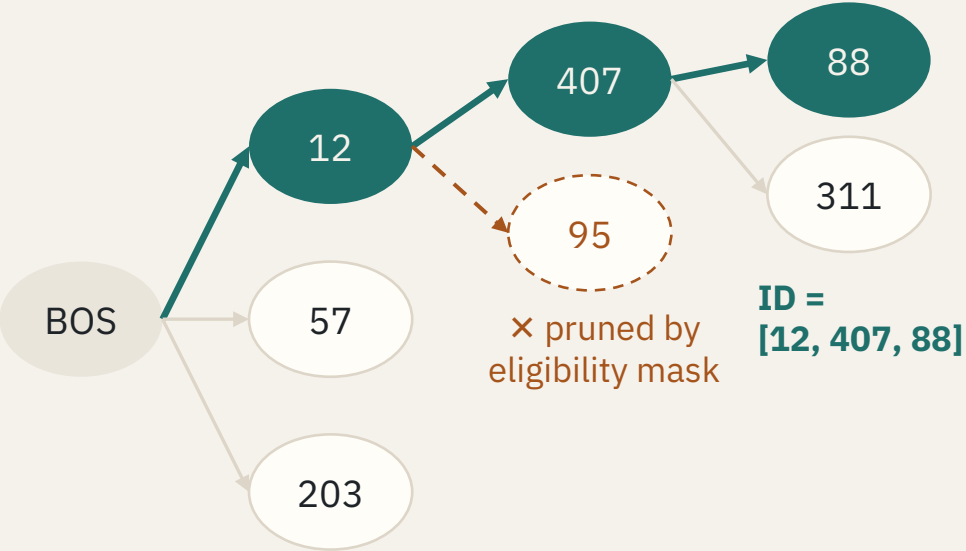
[Figure: Fashion IQ, dataset examples: reference image, relative caption, target image]

Placeholder: PDF not reachable at build time. Wu et al. (2019), arXiv:1905.12794.

Unmentioned attributes **stay fixed** across turns.

Fashion IQ collects relative-caption feedback for dresses, shirts, tops & tees.

Semantic IDs: generate codes, then items



Beam search over codes; highlighted path = one item.

At our 3M items the value is **different candidates**, not efficiency.

[Figure: CQ-SID (Tmall), framework overview]

[Figure: OneSearch (Kuaishou), framework overview]

Placeholder: PDF not reachable at build time. Tmall authors (2026), arXiv:2605.14434.

Placeholder: PDF not reachable at build time. Kuaishou authors (2025), arXiv:2509.03236.

+1.15%

GMV, online A/B (CQ-SID)

72.63%

of purchases via the generative channel (CQ-SID)

-75.40%

operational expenditure (OneSearch)

SEMANTIC ID BY RQ

$$r_0 = e(x), \quad c_\ell = \arg \min_k \|r_{\ell-1} - C_\ell[k]\|, \quad r_\ell = r_{\ell-1} - C_\ell[c_\ell], \quad \text{ID}(x) = (c_1, \dots, c_L)$$

FUSION

RRF is a GAM with hand-fixed shape functions

RECIPROCAL RANK FUSION

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60$$

FUSION AS A GAM (AN EBM GENERALIZES RRF)

$$s(d) = \sum_r f_r(\text{rank}_r(d)) + \sum_{r < r'} f_{rr'}(\text{rank}_r(d), \text{rank}_{r'}(d))$$

An EBM **learns** the shape functions that RRF fixes by hand, plus pairwise terms.

Same rank features in; our L1 already logs them.

RRF CONTRIBUTION, k = 60

rank	1 / (60 + rank)
1	0.0164
10	0.0143
30	0.0111
100	0.0063

Rank 1 counts only 2.6× rank 100: a flat, fixed curve.

Industry moves LLM reasoning off the query path

[Figure: Case-driven multi-agent relevance, framework overview]

Placeholder: PDF not reachable at build time. ByteDance Global E-Commerce Search Relevance Team (2026), arXiv:2605.05991.

[Figure: GenFacet (JD.com), framework overview]

Placeholder: PDF not reachable at build time. Zhai et al. (2026), arXiv:2603.19665.

GenFacet (JD.com): generated facets, +42.0% facet CTR and +2.0% UCVR online.

Case-driven relevance (ByteDance): Annotator, Optimizer and User agents learn from cases.

Our read: the same **compile-once** pattern as RSA.

Our novelty lives in two places:
compiled predicates and persistent
intent state.

Filtering speed, negation scoring and generative retrieval are crowded; we should reuse them.

Where each idea stands against the closest work

Our idea	Closest recent work	Our edge	Gap to close
Compiled predicates	SAE concept features	216 B program on a 56 B code	Beat SAE-derived predicates
Predicates in HNSW	ACORN, FAVOR, EMA	Learned predicate, 96–134 ns	Head-to-head vs ACORN
Negation algebra	E-SENS	Calibrated, composable $\log(1 - p)$	Beat E-SENS on its benchmark
Semantic maps	SAE clustering	LLM-designed, readable axes	Probe accuracy per axis
Intent canvas	Fashion IQ, FACap	Persistent, editable state	Multi-turn edit benchmark
MPDM as a retriever	CQ-SID, OneSearch	Map cells as readable codes	New candidates at 3M items

Stronger rivals to beat next

RSA

ACORN / FAVOR baseline

E-SENS on negation

SAE-derived predicates

Anticorrelated-query slice

Semantic maps

Parametric UMAP and SOM
baselines

Probe accuracy per axis

Axis decorrelation

Intent canvas

Multi-turn Fashion IQ edits

Simulated shoppers

Turns-to-target vs query
rewriting

One-market pilot

Papers in this deck

RSA (our paper)	local PDF	Decoding dense embeddings	arXiv:2506.00041
ACORN	arXiv:2403.04871	Fashion IQ	arXiv:1905.12794
Curator	arXiv:2601.01291	Composed retrieval survey	arXiv:2503.01334
FAVOR	arXiv:2605.07770	FACap	arXiv:2507.07135
EMA	arXiv:2606.00734	CQ-SID (Tmall)	arXiv:2605.14434
Filtered-ANN benchmark	arXiv:2507.21989	OneSearch (Kuaishou)	arXiv:2509.03236
Reproducing NevIR	arXiv:2502.13506	GenFacet (JD.com)	arXiv:2603.19665
E-SENS	arXiv:2608.30130	Case-driven relevance	arXiv:2605.05991
SAE embeddings toolkit	arXiv:2512.10092		